Experiment No. 5

Perform Web Server Reconnaissance Using Various Tools

AIM

To gather information about a target web server, including its software type, version, operating system, and configuration, using reconnaissance tools such as Nmap, Netcraft, and WhatWeb.

PROCEDURE

1. Identify the target web server's domain name or IP address.
2. Use Nmap with service/version detection (nmap -sV -p 80,443) to identify the web server software and open ports.
3. Use the Nmap http-headers and http-server-header NSE scripts to retrieve HTTP response headers.
4. Use the WhatWeb tool to fingerprint the web technologies, CMS, and frameworks running on the server.
5. Use the online Netcraft 'Site Report' tool to obtain the hosting history, server OS, and uptime information of the target.
6. Use a browser or curl -I to manually inspect HTTP response headers (Server, X-Powered-By) for additional clues.
7. Compile the server type, version, OS, and technology stack identified from all sources into a reconnaissance report.

SAMPLE OUTPUT

```
$ nmap -sV -p 80,443 testserver.local
PORT    STATE SERVICE  VERSION
80/tcp  open  http     Apache httpd 2.4.54 ((Ubuntu))
443/tcp open  ssl/http Apache httpd 2.4.54 ((Ubuntu))

$ curl -I http://testserver.local
HTTP/1.1 200 OK
Server: Apache/2.4.54 (Ubuntu)
X-Powered-By: PHP/8.1.2

WhatWeb scan:
  [200] Apache[2.4.54], PHP[8.1.2], WordPress[6.3],
  HTML5, JQuery[3.6.0], Title[Test Server - Home]
```

RESULT

Web server reconnaissance was successfully performed, and the server software (Apache 2.4.54), underlying OS (Ubuntu), scripting language (PHP 8.1.2), and CMS (WordPress 6.3) were identified using Nmap, WhatWeb, and HTTP header analysis.